

Nicolas Monsalves

Research Scientist | Machine Learning & Astronomy (PhD)



Summary

I am passionate about data-driven methods and how to extract answers from the vast amounts of data available today, applying state-of-the-art algorithms to open problems in astronomy. Six years applying deep learning to survey data, astronomical images and simulations on GPU, eight years of scientific software in Python. I am comfortable owning a problem end to end, from raw data and telescope time to a trained model, and working inside large international collaborations.

Contact

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🌐 GitHub
🌐 Portfolio
🌐 LinkedIn

Languages

Spanish (native)
English (fluent)

Technical Skills

Machine learning

Transformers, convolutional neural networks, Balanced Random Forest and more

Frameworks

PyTorch, TensorFlow 2 / Keras, Scikit-Learn, Ray

Engineering

Python (8 years), SQL, Git, Linux/Unix, HPC clusters, LaTeX

Familiar with

Amazon Web Services, Docker

Education

PhD in Astronomy

Universidad de La Serena, 2021 to 2025
With Highest Distinction

Applied Tools for Data-driven Sciences

La Serena School for Data Science, 2021
Student; ten applicants per place

BSc in Astronomy

Universidad de La Serena, 2017 to 2020
Best astronomy graduate of 2021

Research Visits

European Southern Observatory

Garching, Germany
Mar to Jun 2025
Dec 2023 to Jun 2024

NSF NOIRLab

La Serena, Chile, 2023

Certifications

Claude Code in Action and Introduction to Agent Skills
Anthropic, 2026

CNNs in TensorFlow, Intro to TensorFlow for AI, ML, DL, Neural Networks and Deep Learning
DeepLearning.AI, 2020 to 2021

Research Experience

Postdoctoral Fellow, ERC project BHianca

Oct 2025 - Present

Università degli Studi di Roma Tor Vergata

Rome, Italy

Time-domain deep learning: transformers for the early classification of transients in the Nancy Grace Roman Space Telescope.

VLT Large Programme (QueenB): observing shifts on the multi-messenger search for neutron-star and black-hole mergers, combining optical, near-infrared, X-ray and gamma-ray data to characterise their ejecta.

Rapid-response tooling: developed and maintain a Claude Code skill that turns a raw GCN circular into a single quicklook report with the observational constraints.

PhD Researcher, Machine Learning and Astronomy

2021 - 2025

Universidad de La Serena

La Serena, Chile

New massive stars in the Galaxy: automatic classification of stars from Gaia tabular data with a Balanced Random Forest on a highly unbalanced sample, plus automatic line fitting and spectral classification in the LAMOST spectroscopic survey (first-author A&A paper).

Classification of variable stars in big-data surveys: phase-folded light curves encoded as 32x32 images and classified with an optimized CNN, targeting the ten million objects per night expected from LSST (first-author A&A paper).

Image classification of galaxy morphology: developed the CNN for the automatic detection of stellar streams in simulation images (co-author A&A paper).

Teaching Assistant, three editions

2022 to 2024

La Serena School for Data Science

La Serena, Chile

Teaching: assisted the lecturers of an international, NSF-funded data science school attended by around 30 students per edition, from undergraduate to PhD.

Mentoring: supported the students in applying machine learning to their own research projects.

Ongoing Applied ML Projects (2026)

Characterization of variability in massive stars: Balanced Random Forest over tabular features fused with a CNN over images of the same objects, searching the period space quickly and intelligently by combining deep learning with periodograms.

Foundational LAMOST Transformer: self-supervised pretraining on one-dimensional spectra: masked reconstruction with a noise-aware loss, cross-attention heads for regression and classification, representation learning (VICReg); currently studying how to traverse the geometry of the learned space for fast similarity search.

Observing Experience

Hunting symbiotic stars in the Galaxy: a new, purely photometric technique: Swope telescope, Las Campanas Observatory • 11 nights 2022, 10 nights 2023

Sky brightness measurement campaigns: La Silla, Las Campanas and Cerro Tololo Observatories • 3 nights 2023, 2 nights 2024

Experience with VLT instruments: X-shooter, HAWK-I and FORS2 • Paranal, Chile

Refereed Publications

Astrophotometric search for massive stars in the Milky Way: confronting Random Forest predictions with available spectroscopy.

N. Monsalves, A. Bayo, M. Jaque-Arancibia, J. Bodensteiner et al. • *A&A*, 2025

Application of convolutional neural networks to time-domain astrophysics: 2D image analysis of OGLE light curves.

N. Monsalves, M. Jaque-Arancibia, A. Bayo, P. Sánchez-Sáez et al. • *A&A*, 2024

Stream Automatic Detection with Convolutional Neural Networks (SAD-CNN).

A. Vera-Casanova, N. Monsalves, F. A. Gómez, M. Jaque-Arancibia et al. • *A&A*, 2025

A search for new symbiotic stars in the Milky Way.

V. Contreras Rojas, M. Jaque-Arancibia, C. E. Ferreira Lopes, N. Monsalves et al. • *A&A*, 2026

Stellar atmospheric parameters and chemical abundances of ~ 5 million stars from S-PLUS multi-band photometry.

C. E. Ferreira Lopes, L. A. Gutiérrez-Soto, V. S. Ferreira Alberice, N. Monsalves et al. • *A&A*, 2025

A statistical study of lopsided galaxies using Random Forests.

V. Fontirroig, F. A. Gómez, M. Jaque-Arancibia, A. Dolfi, N. Monsalves • *A&A*, 2025

Invited Talks and Seminars

Revealing massive star populations: harnessing big data for deeper understanding

Astronomy Seminars • *Universidad de La Serena, Chile, 2024*

Application of convolutional neural networks to time-domain astrophysics

Optical Gravitational Lensing Experiment • *Warsaw, Poland, 2024*

Studying stellar variability in the big data scenario

INCT Monthly Seminars • *Universidad de Atacama, Chile, 2023*

Contributed Talks

Identifying kilonovae in wide-field infrared transient surveys

COSPAR 2026, session *Exploring the Zoo of Transients in the Era of Big-Data Astronomy* • *Florence, Italy, 2026*

Astrophotometric search for massive stars in the Milky Way

Rodolfo Barbá Conference • *ESA, Villafranca, Spain, 2024*

Look different: classifying periodic time series from a computer vision point of view

ESO Informal Discussion • *ESO Garching, Germany, 2024*

2D image analysis of phase-folded light curves using convolutional neural networks on OGLE data

XVIII SOCHIAS Meeting • *Temuco, Chile, 2023*

Grants and Awards

ESO Scientific Support Discretionary Fund (SSDF): selected to carry out the project *Massive and variable, what is there behind?* • *EUR 4.5k, 2025*

Operational expenses for thesis project, ANID • *EUR $\sim 5k$, 2023 and 2024*

International internship grant, Chilean National Agency for Research and Development (ANID) • *EUR $\sim 9k$, 2023*

Support for national internships in innovation and creation, Universidad de Atacama • *2023*

Faculty of Science Award: best astronomy graduate of 2021 • *Universidad de La Serena, 2022*

National PhD Scholarship, ANID • *EUR $\sim 60k$, 2021*

Supervision and Community Service

Co-advisor, two graduate theses

Lopsided galaxies in IllustrisTNG simulations with machine learning; search for new symbiotic stars in the Milky Way • *Universidad de La Serena, 2022 to 2024*

Organizing committee, ESO Artificial Intelligence Forum • *ESO Garching, Germany, 2024*

Local organizing committee, *What was that?* ESO follow-up for transients, variables and Solar System objects in the LSST era • *ESO Garching, Germany, 2024*

Journal club organizer, Department of Astronomy • *Universidad de La Serena, 2024*

Co-advisor, three undergraduate theses

Gaia DR3 variable star catalogue (eclipsing binaries and periods); eccentric eclipsing binaries with TESS; symbiotic star search in photometric databases • *Universidad de La Serena, 2021 to 2023*

Local organizing committee, LSST/Rubin Observatory workshop • *Universidad de La Serena, 2019*